In [2]:
```
import numpy as np
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
import html
```

In [3]:
```
import plotly.express as px
import plotly.io as pio
pio.renderers.default = "notebook"
```

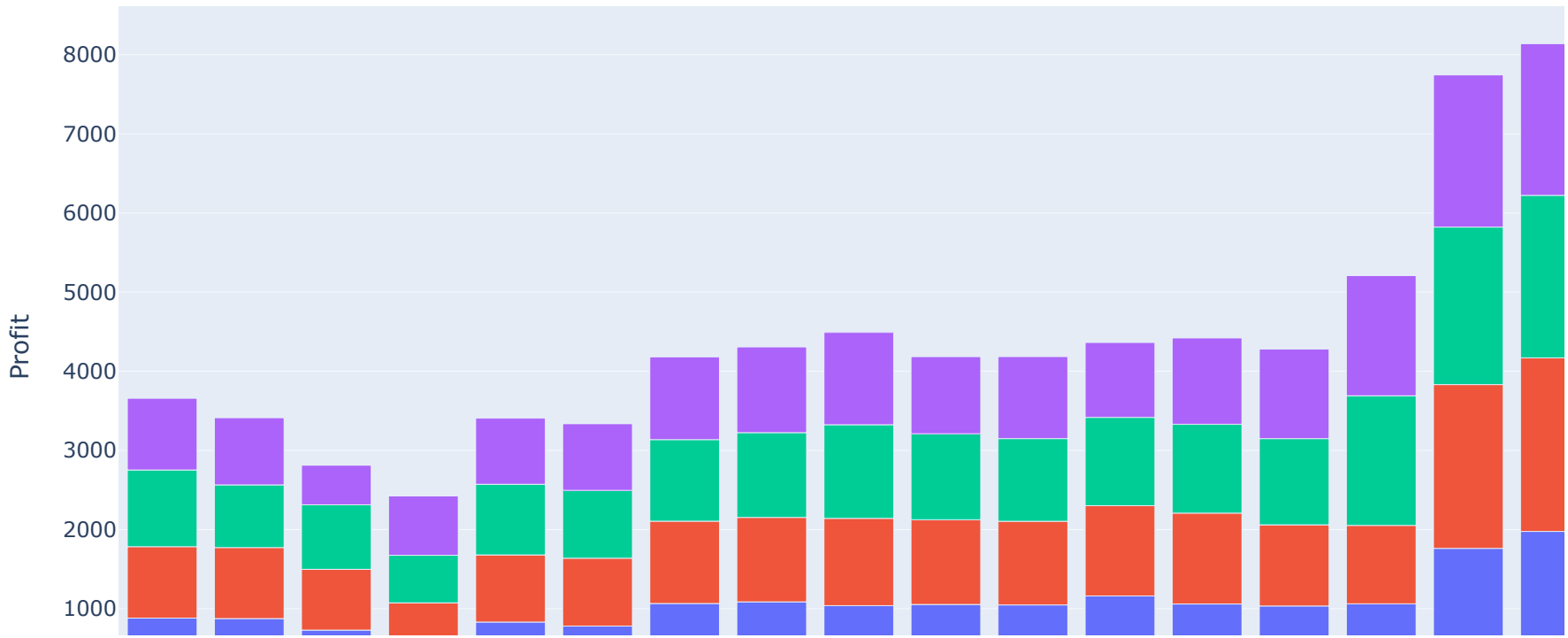
In [4]:
```
fl = pd.read_csv("Data_files\Business_data.csv")
```

In [5]:
```
fl.head()
```

Out[5]:

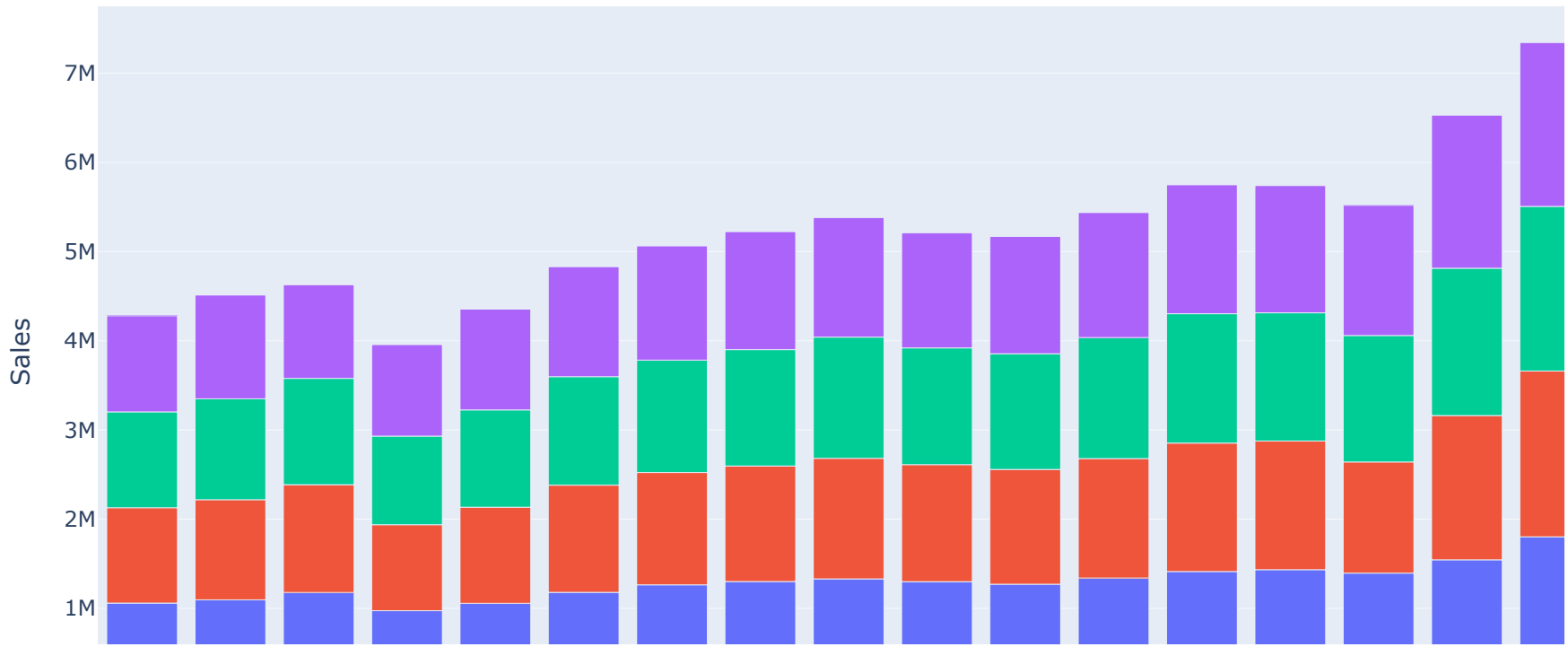
	DATE	N_employees	N_applications	Inflation	BUSINV	Loan_tight	Loan_Demand	Profit	Sales	Quarter	Year
0	1/1/2006	135732.6667	56229.16667	2.504368	1330287.000	-10.7	16.1	880.676	1058565.667	First	2006
1	4/1/2006	136261.6667	55211.53846	2.827660	1362518.667	-12.3	3.5	902.575	1069808.000	Second	2006
2	7/1/2006	136678.3333	47388.57143	3.172047	1389556.000	-8.9	-1.9	969.334	1074451.000	Third	2006
3	10/1/2006	137056.0000	45140.76923	3.181040	1405920.333	0.0	-3.6	905.291	1076955.667	Fourth	2006
4	1/1/2007	137604.3333	54681.53846	3.129683	1417512.333	0.0	-1.8	874.867	1095280.667	First	2007

In [6]:
```
fig_1 = px.bar(fl,x='Year',y='Profit', color='Quarter')
fig_1.show()
```



Looking at the graph showing the profits per quarter for every year since 2006, we can see that the total profit level stayed within a specific range from about 700,000 to about 1.1 million. The graph also shows a dip in profits during the 2008 crash. Unlike the 2008 crash, the covid pandemic shows a rise in profit, mainly due to the third and fourth quarter. So why is this the case. This could be due to an increase in sales levels. But these sales haven't seen much change during this same period.

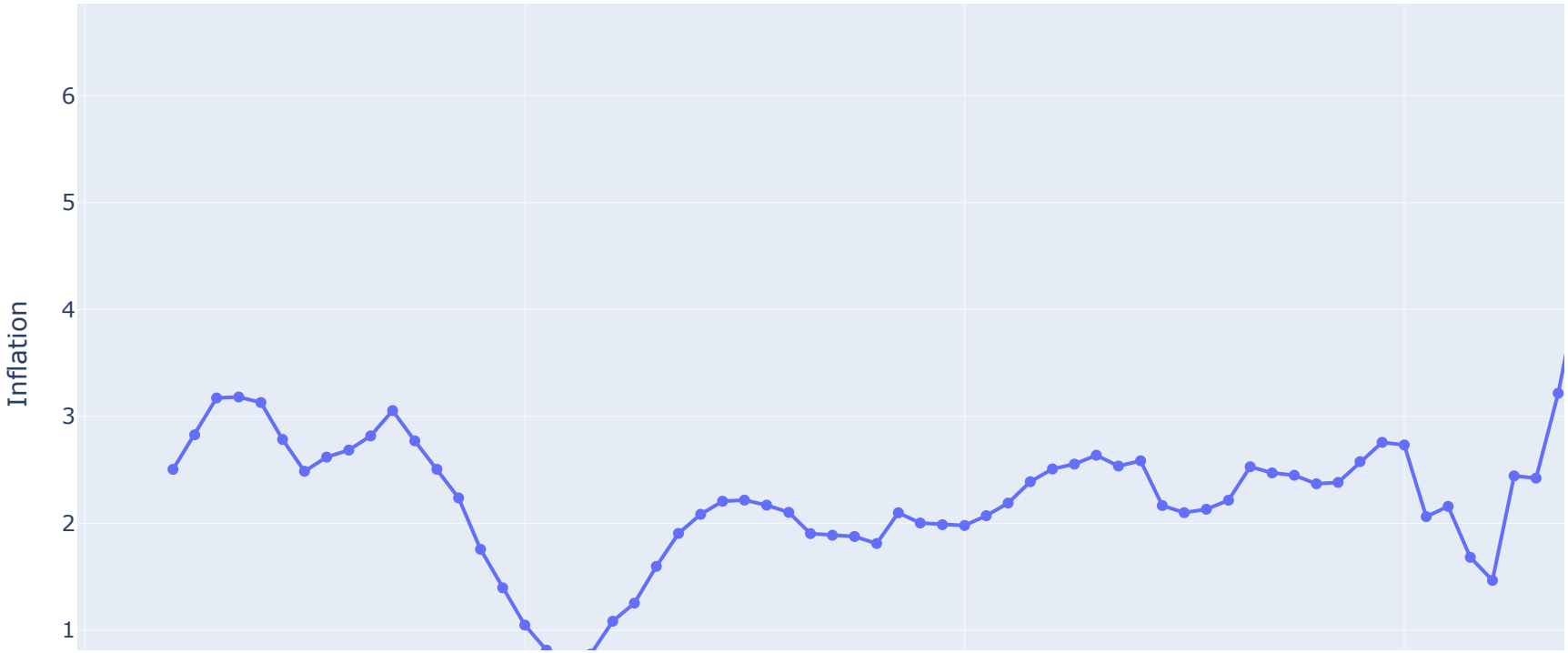
In [7]:
```
fig_2 = px.bar(fl,x='Year',y='Sales', color='Quarter')
fig_2.show()
```



Looking at the graph we can see a general increase in the sales volume with time. But from 2022 to 2023 we can see that there was no increase in the sales, suggesting that we may be soon entering a reccession as the economy may be slowing down. Also we can see a significant increase in the rate of growth of Sales from 2020 to 2021 suggesting that government policies were succesful at bringing out the US economy of the economic recession.

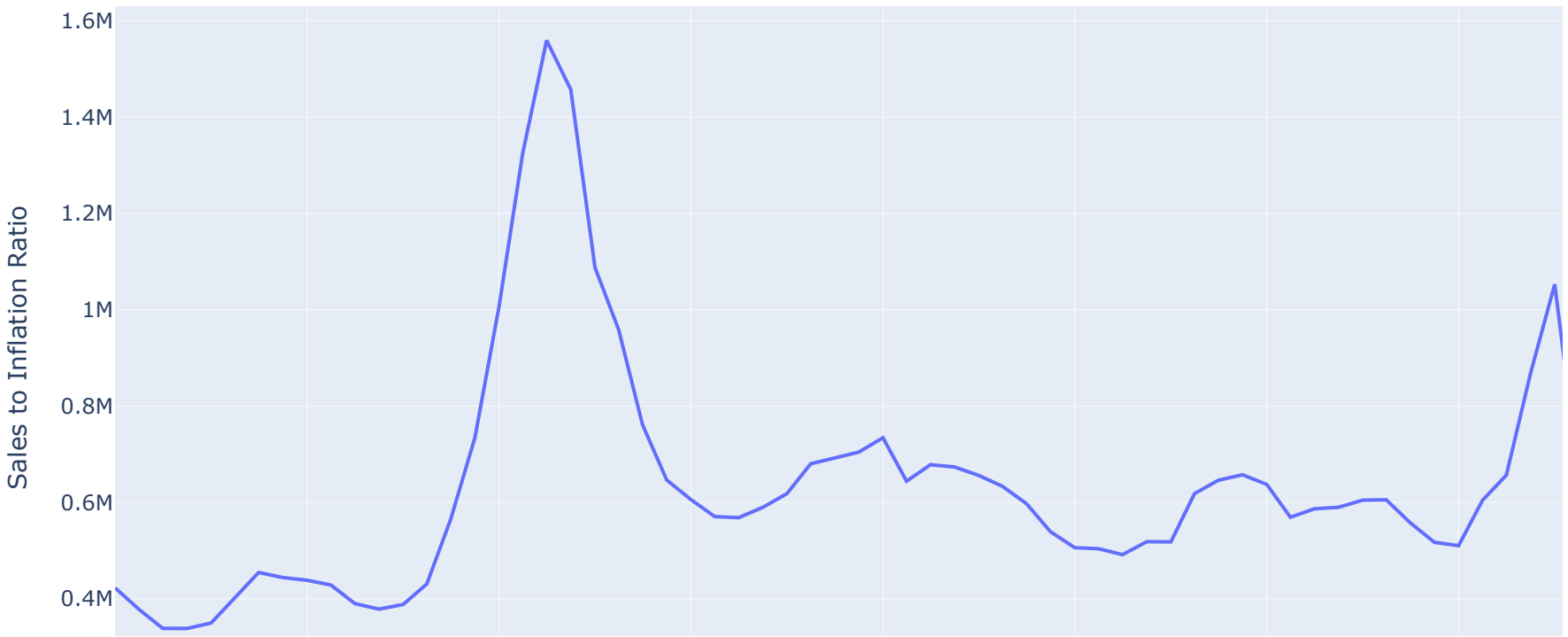
In [8]:
```
fl['DATE'] = pd.to_datetime(fl['DATE'])
fig_3 = px.line(fl,x='DATE',y='Inflation',markers=True)
```

```
fig_3
```

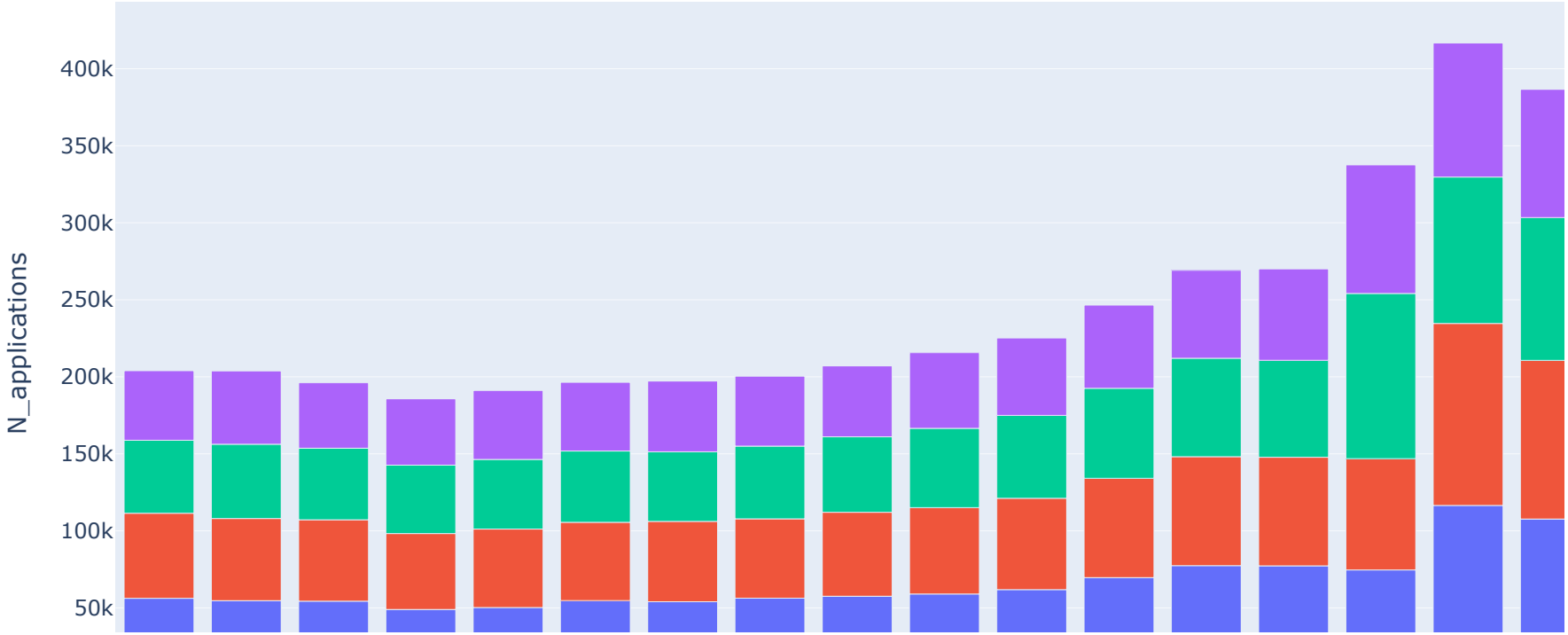


From the graph we notice steeps drops in inflation over periods whereby there were reccessions. The 2008 recession and the 2020 recession. But right after the 2020 recession, we see that inflation rates have gone up significantly, higher than it has ever been in the past 10 years.

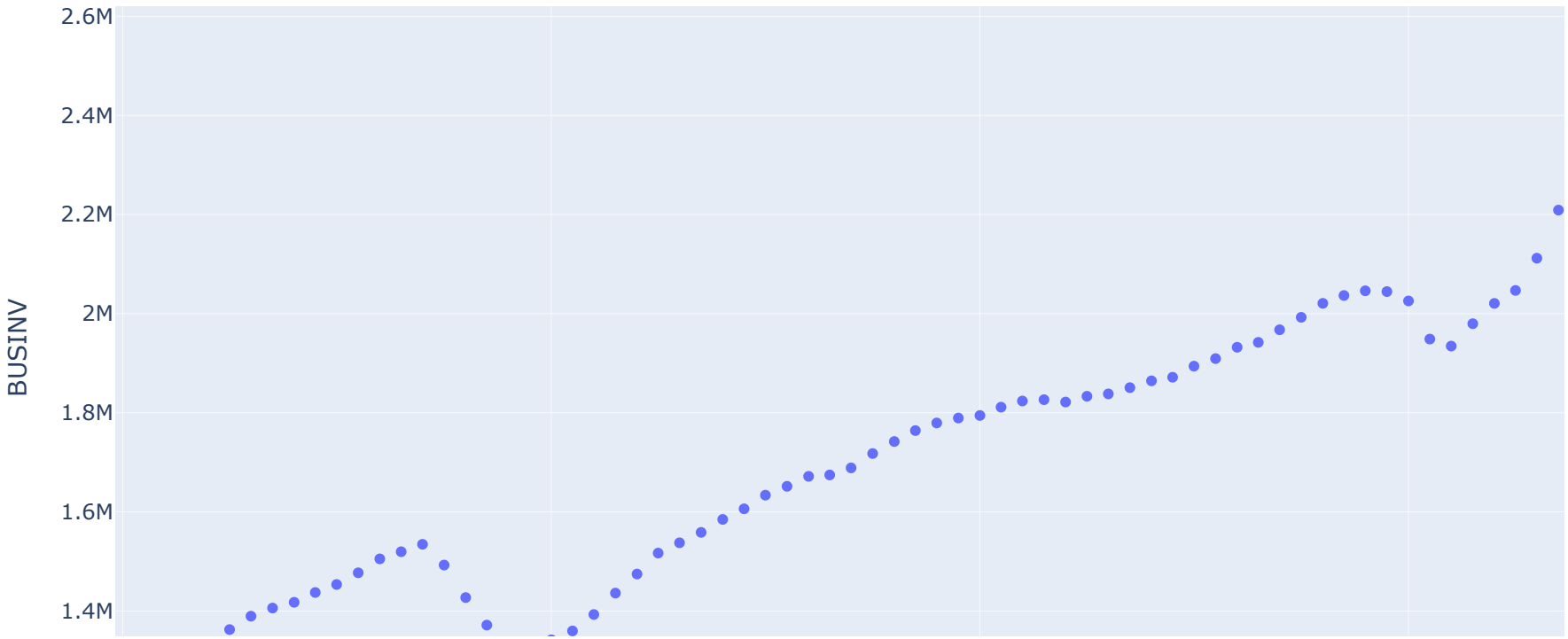
In [9]:
```
px.line(x=fl['DATE'],y=fl['Sales']/fl['Inflation'],labels={
    'y':'Sales to Inflation Ratio','x':'Year','color':'Quarter'
})
```



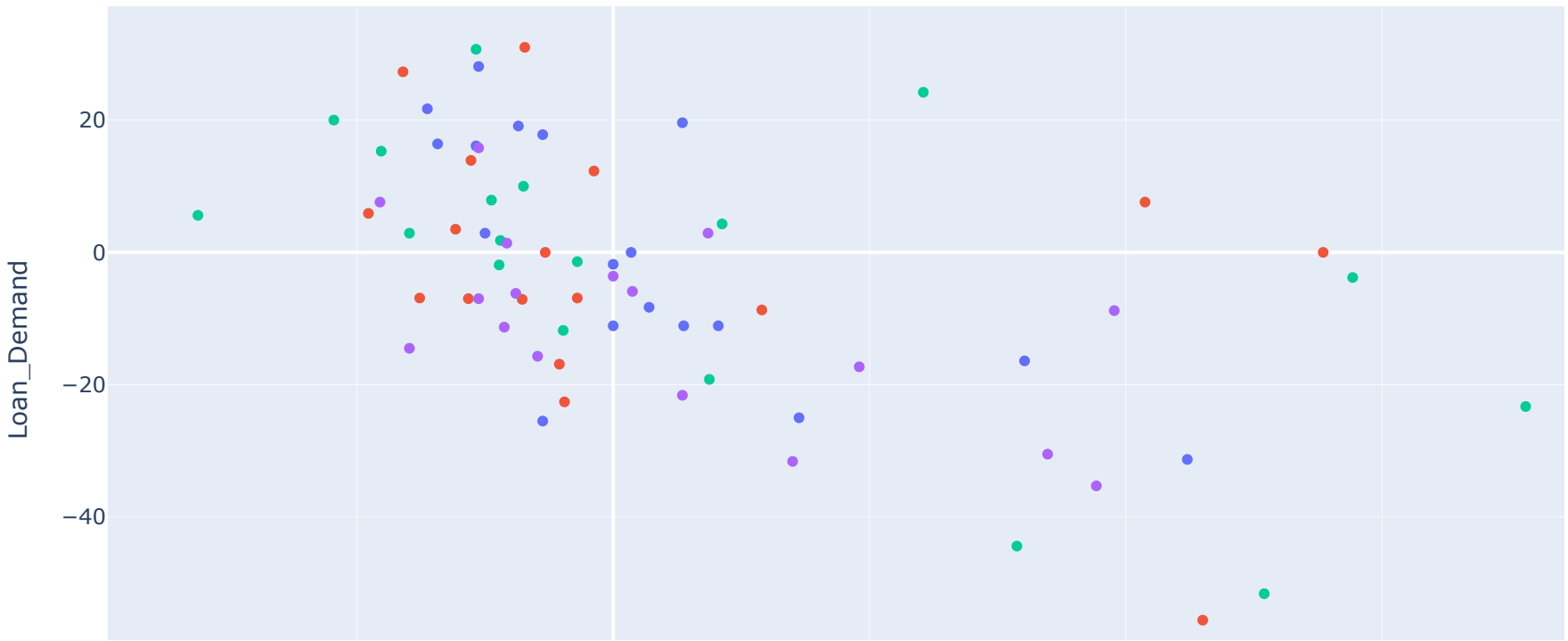
In [10]:
```
fig_4 = px.bar(fl,x='Year',y='N_applications', color='Quarter')
fig_4.show()
```



In [11]:
```
px.scatter(fl,x='DATE',y='BUSINV')
```



In [12]:
```
px.scatter(fl, x='Loan_tight', y='Loan_Demand', color='Quarter')
```



In [13]:
```
import sklearn
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
```

In [14]:
```
'''st = pd.DataFrame()
st['BUSINV'] =(fl['BUSINV']-fl['BUSINV'].mean())/fl['BUSINV'].std()
st['N_employees'] =(fl['N_employees']-fl['N_employees'].mean())/fl['N_employees'].std()
st['N_applications'] =(fl['N_applications']-fl['N_applications'].mean())/fl['N_applications'].std()
```

```
st['Inflation'] =(fl['Inflation']-fl['Inflation'].mean())/fl['Inflation'].std()
st['Loan_tight'] =( fl['Loan_tight']-fl['Loan_tight'].mean())/fl['Loan_tight'].std()
st['Loan_Demand'] =(fl['Loan_Demand']-fl['Loan_Demand'].mean())/fl['Loan_Demand'].std()
st['Profit'] =( fl['Profit']-fl['Profit'].mean() )/fl['Profit'].std()
st['Sales'] =(fl['Sales']-fl['Sales'].mean())/fl['Sales'].std()'''
#The above code is manual standardization
```

Out[14]:

```
"st = pd.DataFrame()\nst['BUSINV'] =(fl['BUSINV']-fl['BUSINV'].mean())/fl['BUSINV'].std()\nst['N_employees'] =(fl['N_employee
s']-fl['N_employees'].mean())/fl['N_employees'].std()\nst['N_applications'] =(fl['N_applications']-fl['N_applications'].mean
())/fl['N_applications'].std()\nst['Inflation'] =(fl['Inflation']-fl['Inflation'].mean())/fl['Inflation'].std()\nst['Loan_tig
ht'] =( fl['Loan_tight']-fl['Loan_tight'].mean())/fl['Loan_tight'].std()\nst['Loan_Demand'] =(fl['Loan_Demand']-fl['Loan_Dema
nd'].mean())/fl['Loan_Demand'].std()\nst['Profit'] =( fl['Profit']-fl['Profit'].mean() )/fl['Profit'].std()\nst['Sales'] =(fl
['Sales']-fl['Sales'].mean())/fl['Sales'].std()"
```

In [15]:

```
pc = fl.loc[:72,'N_employees':'Sales']
dat2 =StandardScaler().fit_transform(pc.loc[:, pc.columns!='Inflation'])
pca = PCA(0.95)
data_pca = pca.fit_transform(dat2)
data_pca.shape
```

Out[15]:

```
(73, 4)
```

Therefore our data has been reduced to 4 components which explains 95% of the variance.

In [16]:

```
pca.explained_variance_ratio_
```

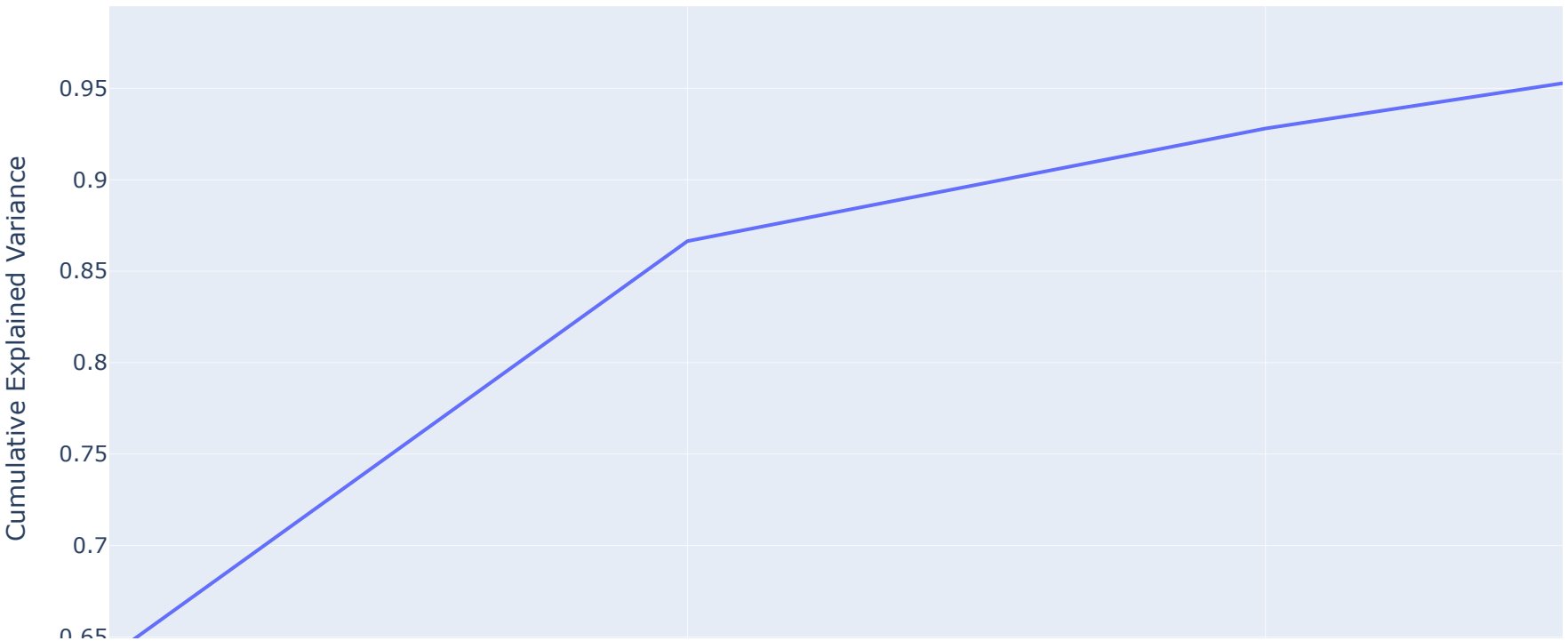
Out[16]:

```
array([0.63914475, 0.22729467, 0.06159586, 0.04810518])
```

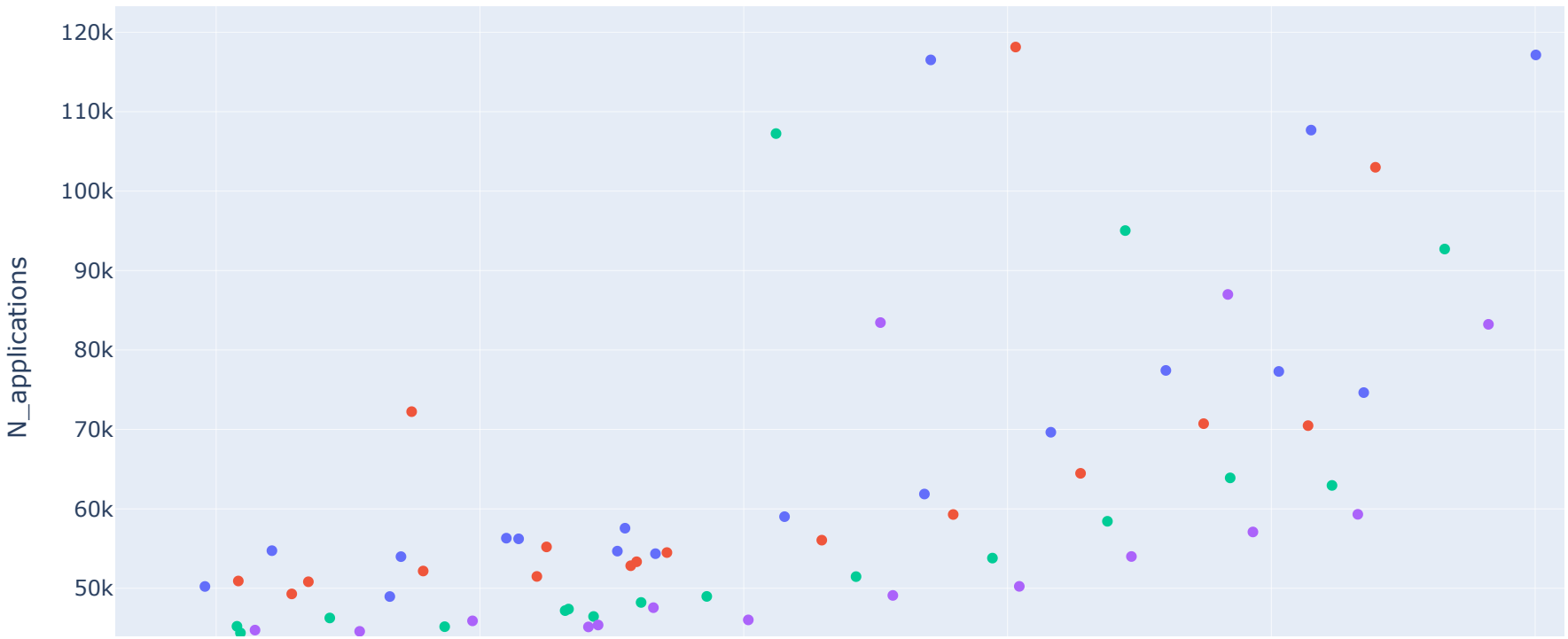
This is the explained variance for each component, component 1 holds a significant portion of the variance.

In [17]:

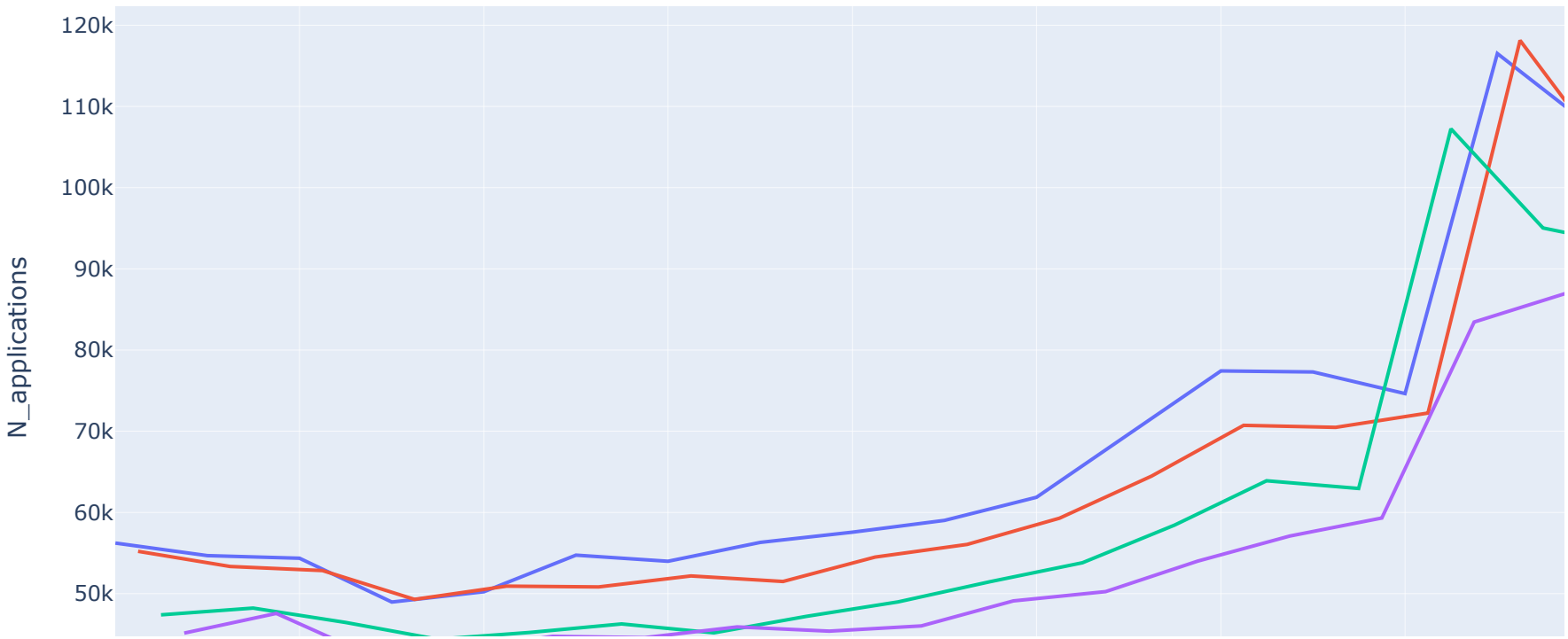
```
fig_5 =px.line(y=np.cumsum(pca.explained_variance_ratio_), x=[1,2,3,4], labels={
    'y':'Cumulative Explained Variance', 'x':'Number of components'
},)
fig_5.update_xaxes( tick0 = 1, dtick = 1)
```



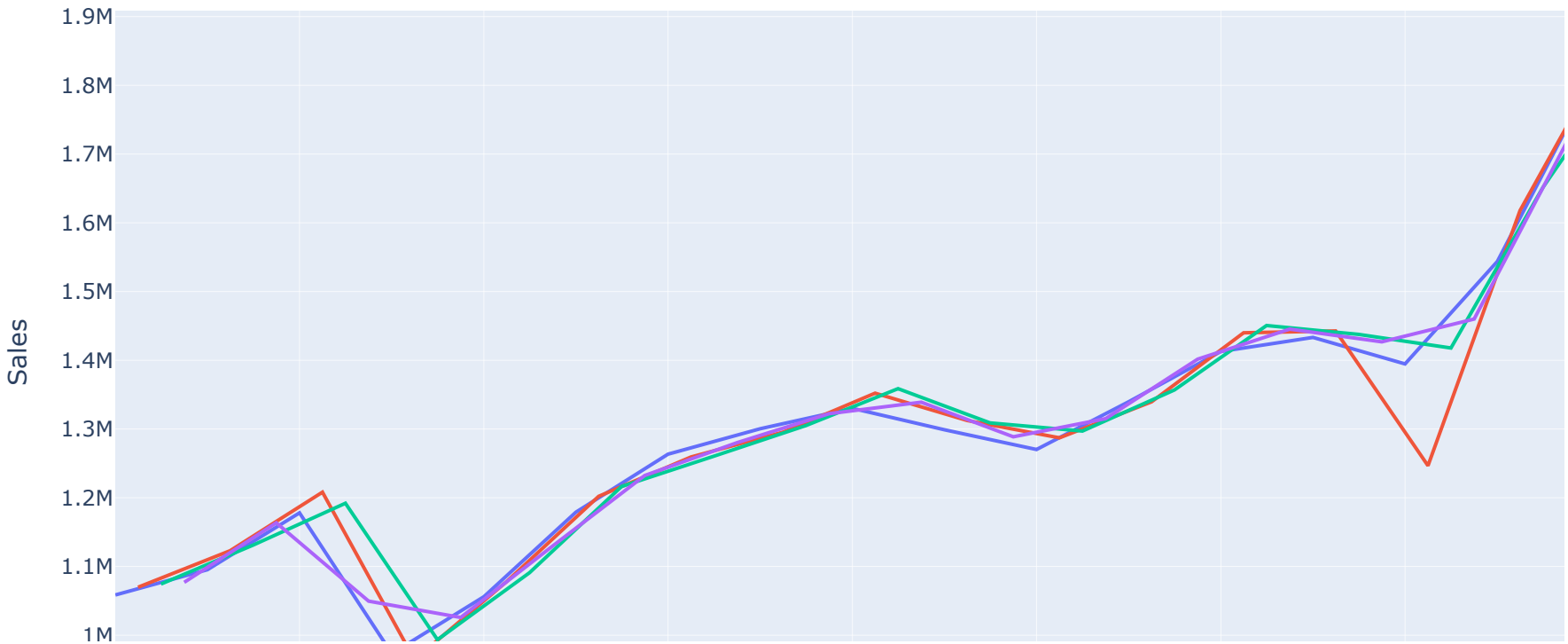
In [18]:
```
px.scatter(fl, x='N_employees', y='N_applications', color='Quarter')
```



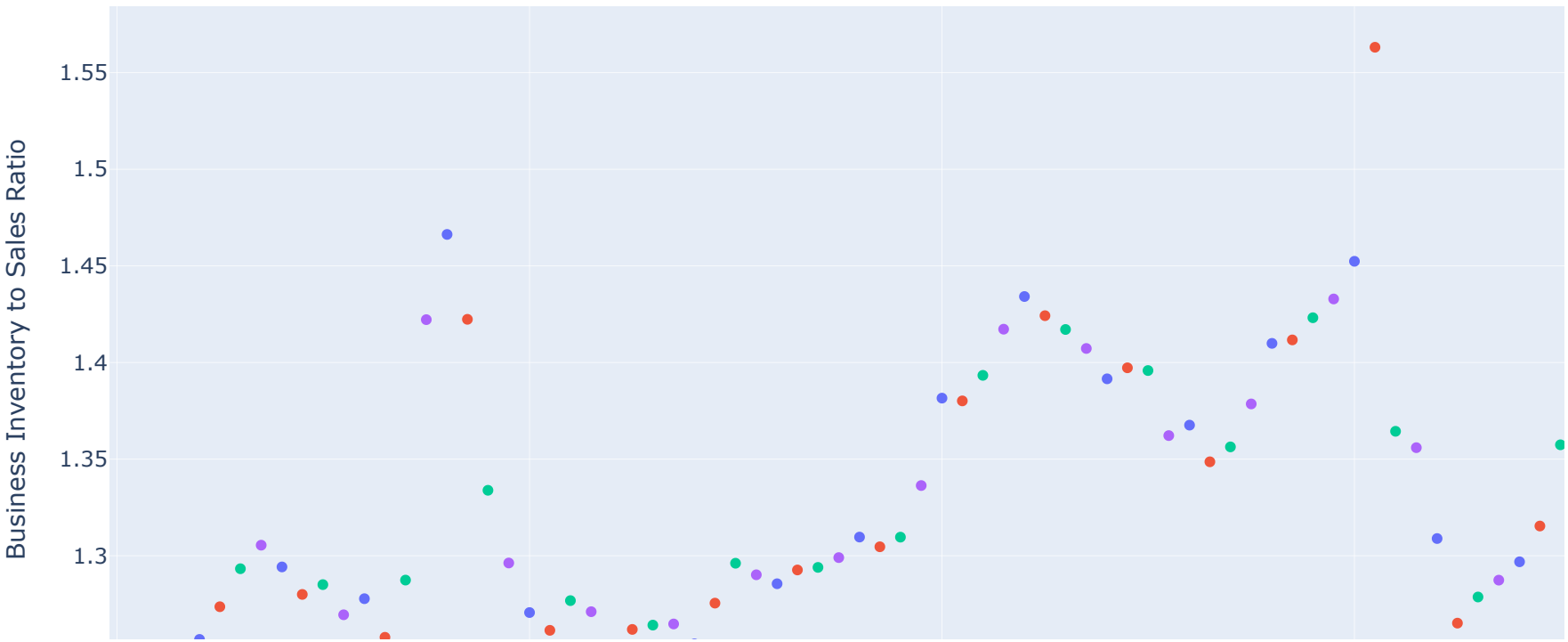
In [19]:
```
px.line(fl,x='DATE',y='N_applications',color='Quarter')
```



In [20]:
```
px.line(fl,x='DATE',y='Sales',color='Quarter')
```



In [21]:
```
px.scatter(x=fl['DATE'],y=fl['BUSINV']/fl['Sales'], color=fl['Quarter'],labels={
    'y':'Business Inventory to Sales Ratio','x':'Year','color':'Quarter'
})
```


Business Inventory to Sales Ratio
1.55
1.5
1.45
1.4
1.35
1.3